

Exam Date & Time: 17-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

DATA ARCHITECTURE [DS_3151]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A national energy company, "PowerGrid," collects two primary data types: (1) Billions of smart meter readings and (2) Thousands of customer complaints left as free-form notes in the CRM.
 - A) Critically evaluate the difference in processing strategy between these two data streams based on the "Great Divide". (5)

Explain why processing the meter readings is considered Hadoop-centric and relies on parsing being "almost trivial," whereas the customer notes must emphasize Textual Disambiguation.
 - B) Analyze the role of Taxonomies in the analysis of the customer complaint notes. Explain how a taxonomy allows PowerGrid to unify queries on a broad concept (e.g., "service outage") even if only specific instances (e.g., "power cut," "cable down") appear in the original text. (3)
 - C) Apply the concept of Custom Variables or Named Value Processing to describe how PowerGrid could automatically identify and tag a standard customer ID format consistently embedded within the customer service notes based purely on format recognition. (2)
- 2) A government agency is modernizing its systems. They are debating whether to use the artifact-centric Zachman Framework or the process-centric TOGAF Architecture Development Method (ADM)
 - A) for guiding the modernization initiative. (5)

Critically evaluate the difference in focus between the process-oriented TOGAF ADM (as an iterative process, Phase A-H) and the artifact classification grid of the Zachman Framework. Which framework is generally better suited for achieving rapid business value?
 - B) Analyze the architectural guiding principles: Evolution of the Object (extensibility) and designing for Current and Future Use (repurposing). Explain why incorporating these principles is vital for building systems that remain agile as business needs mature. (3)
 - C) Explain the purpose of Phase A: Architecture Vision in this context. Describe how it (2)

helps the team establish the project scope, identify key stakeholders, and gain management approval before moving to the detailed design phases.

- 3) NextGen's logical model includes complex entities like Employee that relate to themselves, and requires detailed historical tracking for critical changes.
 - A) Critically evaluate the necessity of Complex History (Historical Logging) over Simple or Bounded Simple History. Justify why Complex History is mandatory when the integrity of the audit trail requires the capability to handle retroactive or out-of-sequence changes. (5)
 - B) Analyze the problem of Recursive Relationships and how this structure can lead to Duplicate Propagated Keys. Explain the resolution techniques needed: renaming the key or selectively removing the identifying nature of the relationship. (3)
 - C) Apply the concept of history by defining the implementation of Bounded Simple History. Specifically, identify the two time-dependent keys required for tracking a non-critical attribute change. (2)
- 4) FinTech manages its designs in a Model Repository. The logical model must be transformed into a Physical Data Model (PDM) for an OLTP system, which must remain separate from the Data Warehouse (DW).
 - A) Evaluate the necessity of a strict Model Repository Policy that governs Subsetting and Reconciliation . Explain why reconciliation is crucial for reintegrating the application PDM back into the Enterprise Logical Model. (5)
 - B) Analyze why OLTP Data Stores are fundamentally unsuitable for being used directly as a Data Warehouse. Focus your analysis on the conflict between optimizing for transactional performance versus high-volume DSS retrieval, and the requirement for noncontention. (3)
 - C) Define the physical design concept of Horizontal Segmentation. Explain when this technique is typically applied, using data volume, access patterns, or archiving needs as justification (2)
- 5) The physical design team at FinTech Solutions is conducting a final design review for its Customer Account Management System. They are using Interaction Analysis tools including the CRUD Matrix and the Entity Life Cycle Analysis (ELCA), to validate whether all business processes and data entities are correctly aligned.
 - A) Evaluate how these two tools complement each other in verifying the completeness and correctness of the system's design. Specifically, explain how the ELCA ensures that every entity (e.g., Customer Account) transitions through all valid states, while the CRUD Matrix validates that every process properly performs the required data operations on the corresponding entities. (5)

- B) The database designer uses a Composite Load Map to visualize how multiple system processes access shared data tables.

Analyze how this map helps identify Hot Spots (tables or columns accessed too frequently) and define what Referential Groups and Unit of Work Constraints reveal about the system's data access behavior. (3)

- C) In a reporting database, too many small tables are slowing down queries.

Explain how Entity Consolidation can simplify the schema and improve query performance. (2)
Also describe one drawback of Overnormalisation that might lead to excessive joins and loss of practical "data intelligence."

-----End-----

Event Name		Subject	
058-END SEMSTER-ODD-2025		DS_3151:DATA ARCHITECTURE	
#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	15
4	4	CO 4	15
Total			50

* NA - CO Not Selected

Event Name		Subject	
058-END SEMSTER-ODD-2025		DS_3151:DATA ARCHITECTURE	
#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	5	10(20%)
4	Analysis	5	15(30%)
5	Create	5	25(50%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)